Reviewer-Level Research Roadmap for a WiFi CSI Gait Recognition Dataset

Clean formatted version of the Pro model response

Source: uploaded Pro response document

I'll ground the recommendations in your uploaded dataset description and recent CSI/WiFi sensing literature, separating publishable novelty from baseline work to build concrete Q1 paper plans.

A first reviewer-level read is that the regular-router data is usable for robustness-focused CSI gait work, while the modified-router subset should be framed as cross-setup evidence rather than a standalone hardware benchmark.

Executive diagnosis

Your dataset is **publishable**, but it is not automatically Q1-ready. Its strongest value is not simply "25 people with WiFi CSI gait samples." The stronger story is:

A WiFi CSI gait/person-recognition dataset with controlled direction labels, carrying-condition variation, and a small paired regular-vs-antenna-spread router subset that can support robustness, domain-shift, and representation-learning studies.

From your description, the regular setup has **25 participants × 100 samples = 2,500 samples**, and the modified setup has **5 overlapping participants × 100 samples = 500 samples**. Each sample has a CSI file and CSV file. Odd/even sample indices encode walking direction, and sample ranges 1–80 versus 81–100 encode normal walking versus bag-carrying walking. The modified-router subset uses the same 5 people who also appear in the regular setup.

This gives you four useful label axes:

Axis	Labels	Strength
Person ID	25 regular, 5 modified-overlap	Main biometric task
Direction	left-to-right / right-to-left	Strong robustness task
Carrying condition	normal / with bag	Strong covariate-shift task, but imbalanced
Setup/domain	regular / modified antenna-spread	Novel, but only 5 subjects

The **most realistic Q1 route** is not a simple accuracy paper. A paper that says "we trained CNN/LSTM/Transformer and achieved high accuracy on 25 people" will look incremental. A Q1-level paper must emphasize **hard evaluation protocols**, **anti-leakage splitting**, **cross-direction/cross-bag generalization**, **domain adaptation**, **few-shot/metric learning**, **statistical reliability**, and **explainability**.

PART 1 — Literature positioning

1.1 Where the field is now

WiFi CSI sensing is now a mature research area. CSI captures fine-grained propagation changes, including multipath, attenuation, and phase shifts caused by moving bodies; recent surveys explicitly classify WiFi CSI sensing into applications such as activity recognition, gesture recognition, vital signs, gait analysis, and people identification.

Early gait/person-recognition systems such as **WiWho** showed that CSI-based gait can identify people device-free. WiWho evaluated 20 volunteers and reported average identification accuracy decreasing from about 92% to 80% as group size increased from 2 to 6 people. It also explicitly positioned WiFi gait ID as suitable for smart-space convenience rather than high-risk biometric authentication.

WiFi-ID similarly demonstrated that walking-induced CSI perturbations can identify people, reporting 93% to 77% accuracy for group sizes from 2 to 6 people, using time/frequency features and feature selection.

Later work moved toward deep learning and more realistic covariates. **Gate-ID** is especially relevant to your dataset because it targets direction-invariant identification. It used a WiFi "virtual gate," studied mirror-like CSI patterns caused by opposite walking directions and antenna orientation, used spatial-temporal features plus augmentation, and evaluated 20 subjects crossing a WiFi transceiver pair back and forth.

More recent papers emphasize robustness, transfer learning, few-shot learning, multimodality, and domain adaptation. **CAUTION** introduced few-shot open-set WiFi human authentication and uses low-dimensional user profiles plus an intruder threshold so the system can work with limited CSI training data and reject unknown users. **AutoFi** introduced geometric self-supervised learning for annotation-efficient WiFi sensing and transfers knowledge from randomly captured CSI samples to tasks including gait recognition. **DCS-Gait** is directly relevant to your modified-router subset: it studies cross-scene and cross-state WiFi CSI gait recognition, aligns class-level distributions using a cross-attention metric, and evaluates 34 subjects across 2 scenes and 3 states.

Transformer-style and hybrid deep models are now common. **Wi-Gait** uses a multiscale residual convolution-enhanced Transformer encoder to learn more robust gait features under walking cofactors and device-placement variation. **GaitFi** combines WiFi CSI with vision, uses a lightweight residual convolution network, triplet loss plus classification loss, and reports 94.2% human-identification accuracy for 12 subjects. **NeuralWiGait** uses CNN + BiGRU and reports high accuracy on public datasets such as Widar 3.0 and NTU-Fi-HumanID.

The field is also becoming stricter about evaluation. Recent work has shown that WiFi CSI human-sensing papers can suffer serious **data leakage** when windows or samples from the same subject/session are split incorrectly; one Sensors 2024 study highlights inflated performance caused by absence of subject-based partitioning. This matters directly for your dataset because you have 100 samples per participant collected in a common environment.

1.2 Representative comparison with your dataset

Work / dataset	Typical focus	Participants / scale	Key method direction	Limitation your dataset can address
WiWho	Device-free person ID from CSI gait	20 volunteers; small groups	Step/walk analysis, handcrafted gait features	Limited group size, less modern DL, limited covariates
WiFi-ID	Human identification using CSI perturbations	20 subjects; group sizes 2–6	Time/frequency features, feature selection	Accuracy drops with group size; weak modern robustness
Gate-ID	Direction-invariant ID	20 subjects	Direction estimation, feature augmentation, attention model	Focused virtual-gate setup; your dataset has explicit odd/even direction labels

Work / dataset	Typical focus	Participants / scale	Key method direction	Limitation your dataset can address
Widar/GaitID-related gait dataset	Cross-domain WiFi sensing and gait	Gait dataset reported as 22K instances from 11 participants	BVP/gait-specific features and transfer ideas	Your dataset has more identities than the GaitID subset but fewer total instances
CAUTION / NTU-HumanID	Few-shot open-set human authentication	Benchmark lists 14 gait classes in NTU-HumanID	Few-shot profiles, intruder threshold	Your 25-subject regular subset can support better closed-set and metric-learning studies
DCS-Gait	Cross-scene/cross-state domain adaptation	34 subjects, 2 scenes, 3 states	Class-level domain adaptation	Your 5-person setup-transfer subset is interesting but too small alone
GaitFi	WiFi + vision multimodal ID	12 subjects	LRCN, triplet + classification loss	You have WiFi-only, so your contribution must be robustness/protocol/modeling, not multimodal advantage
CSI-Bench / modern benchmarks	Real-world multitask WiFi sensing	35 users, 26 environments, 461 hours	Standardized splits, multitask baselines	Shows why Q1 reviewers now expect generalization protocols and reproducibility

Your dataset is **stronger than many early WiFi gait ID works in participant count**, but **weaker than the newest cross-environment benchmarks** in environmental diversity, session diversity, and public benchmarking scale.

PART 2 — Deep dataset analysis

2.1 What makes the dataset strong

First, **25 participants** is respectable for WiFi CSI gait/person recognition. It is larger than many early gait-ID demonstrations and larger than the 11-participant gait subset associated with GaitID/Widar reporting.

Second, each participant has **100 repeated walking samples**, which gives enough repeated trials for deep learning, cross-validation, and reliability testing. You have 2,500 regular-router samples and 500 modified-router samples, assuming no missing files.

Third, the dataset has **built-in covariate labels**: direction and carrying condition. Many CSI gait papers treat direction, speed, trajectory, or carrying condition as nuisance variation. Your indexing rule gives an immediate way to test robustness rather than only reporting easy random-split accuracy.

Fourth, the **modified-router antenna-spread subset** is genuinely interesting. Detaching two 5 GHz antennas and separating them by roughly 4 m creates a different spatial illumination pattern and likely changes Fresnel-zone coverage, multipath diversity, phase relationships, and body-shadowing sensitivity. This can be framed as a **low-cost spatial aperture / antenna-placement intervention**, but only carefully.

Fifth, the same 5 participants appear in both regular and modified setups, giving a paired cross-domain subset. That enables regular → modified and modified → regular adaptation experiments.

2.2 What makes the dataset weak

The main weakness is **single-environment bias** if all data were collected in one room/corridor. CSI models often learn environment-specific multipath. Without multiple rooms, reviewers may say the model is recognizing a person-in-room signature rather than portable gait biometrics.

The second weakness is that the modified-router subset has only **5 participants**. That is useful for pilot cross-setup evaluation, but it is not enough for a standalone high-impact claim about antenna-spread router design.

The third weakness is **bag-condition imbalance**: 80 normal and 20 bag samples per person. This is acceptable for robustness testing, but weak for a primary bag/no-bag classification claim unless you use balanced metrics and carefully controlled splits.

The fourth weakness is possible **session leakage**. If the 100 samples per participant were collected continuously, random splits may put nearly identical temporal/environmental samples into train and test.

The fifth weakness is that you have not yet documented metadata that Q1 reviewers will expect: room geometry, AP/RX position, antenna height/orientation, walking path, channel, bandwidth, packet rate, number of subcarriers, packet drops, participant demographics, bag weight, collection session, and environmental conditions.

2.3 Missing metadata that may hurt Q1 acceptance

You should document these before model training:

Metadata category	Required details
Hardware	Intel AX210 receiver, TP-Link Archer C6 v2 AP, antenna count, antenna orientation, extension cable length, RF-chain mapping
WiFi PHY	Band, center frequency/channel, bandwidth, 802.11 mode, MCS, NSS, guard interval, transmit power, AP beacon/traffic generation method
PicoScenes	version, command, AP mode settings, exported fields, CSI dimension, timestamp format
Geometry	room size, AP-RX distance, walking path length, path coordinates, line-of-sight / non-line-of-sight, antenna heights

Metadata category	Required details
Modified setup	exact 4 m antenna coordinates, cable type/length, whether only two antennas were moved, whether impedance/cable loss was controlled
Participant	anonymized ID, age range, sex, height, weight, shoe type if available, dominant walking style notes
Condition	bag type, approximate bag weight, carrying side, whether participants carried it consistently
Collection process	sample order, breaks, collection date/time, who was present, background motion, WiFi interference
File integrity	packet count, duration, dropped-packet rate, missing CSI files, corrupted CSV files
Ethics	consent, anonymization, intended use, biometric-risk disclosure

Without these, the paper can still be submitted, but reviewers may call it insufficiently reproducible.

2.4 Research questions your dataset can answer

Your dataset can answer:

1. **Can WiFi CSI identify 25 known people from gait in a fixed indoor setup?**
2. **How much does walking direction affect CSI-based person recognition?**
3. **Can a model trained on left-to-right walking generalize to right-to-left walking?**
4. **How much does carrying a bag degrade biometric recognition?**
5. **Can multi-task learning improve person recognition by jointly learning direction and carrying condition?**
6. **Can self-supervised pretraining reduce the number of labeled samples required per person?**
7. **Can metric learning support few-shot enrollment for unseen or low-sample identities?**
8. **Does the modified antenna-spread router create more discriminative CSI signatures than the regular router for the same 5 people?**
9. **Can domain adaptation transfer recognition from regular-router CSI to modified-router CSI?**
10. **Which CSI representation is most robust: amplitude, phase, amplitude+phase, PCA, spectrogram, or transformer token sequence?**

2.5 Label generation

Use these deterministic labels:

```
participant_id = folder/person identifier
sample_id      = integer parsed from filename
direction:
    if sample_id % 2 == 1: left_to_right
    else: right_to_left
condition:
    if 1 ≤ sample_id ≤ 80: normal
    if 81 ≤ sample_id ≤ 100: with_bag
setup_type:
  regular or modified
domain:
  regular_router or modified_antenna_spread
```

For bag samples, direction still applies:

Range	Odd	Even
1–80	normal + left-to-right	normal + right-to-left
81–100	bag + left-to-right	bag + right-to-left

2.6 Leakage risks

The biggest leakage risks are:

Leakage risk	Why dangerous	Fix
Window leakage	Multiple windows from one CSI sample appear in train and test	Split at sample level before windowing
Participant-session leakage	Random samples from same continuous session appear in train/test	Use block/sample-index splits and hard condition splits
Preprocessing leakage	Normalization/PCA fitted on full dataset	Fit scaler/PCA only on train, apply to val/test
Augmentation leakage	Augmented version of test sample appears in train	Augment only train samples
Setup leakage	Same modified subject samples used in both source and target tuning/test	Separate adaptation, validation, and final target test
Hyperparameter leakage	Test set repeatedly used for model choice	Keep a locked final test protocol
Identity leakage in auxiliary tasks	Direction/bag classifier learns person identity due to imbalance	Use stratified splits and report per-person metrics

Recent CSI-HAR leakage critiques show why this is not a minor issue; subject-unaware or improper partitioning can substantially inflate reported performance.

2.7 Recommended splits

Split A — weak baseline only: random sample split

Use 60/20/20 or 70/15/15 per person, stratified by direction and condition.

This is useful only as an **easy in-distribution reference**. Do not make your main claim from this.

Split B — stronger closed-set person-ID split

For each participant in regular setup:

Set	Normal samples	Bag samples	Direction balance
Train	1–60	optional none or 81–90	balanced odd/even
Val	61–70	optional 91–94	balanced
Test	71–80	95–100	balanced as much as possible

Better: keep **bag samples entirely out of training** for cross-bag robustness.

Split C — cross-direction

Train on left-to-right only, validate on a subset of left-to-right, test on right-to-left. Repeat reversed.

This is strong because direction affects CSI waveform symmetry and antenna orientation, as shown in Gate-ID.

Split D — cross-bag

Train on normal walking samples 1–80, test on bag samples 81–100. This is one of your most important protocols.

Split E — combined hard split

Train on normal left-to-right, test on bag right-to-left. Repeat train normal right-to-left, test bag left-to-right.

This is very strong because both direction and carrying condition shift simultaneously.

Split F — cross-setup transfer

Use only the 5 overlapping subjects:

Protocol	Source	Target
R → M	regular setup	modified setup

Protocol	Source	Target
M → R	modified setup	regular setup
Unsupervised DA	labeled source + unlabeled target train	target test
Few-shot DA	labeled source + k target samples/person	remaining target test

This should be presented as a **domain-shift pilot**, not as definitive proof of deployment generalization.

Split G — leave-one-person-out / unseen-person protocol

Closed-set classification cannot directly use leave-one-person-out because the test person is not in the classifier's output classes. Use this only for:

- verification,
- open-set recognition,
- few-shot enrollment,
- metric learning,
- representation evaluation.

Example: train an embedding on 24 people; for the left-out person, enroll k samples and query the rest. Report EER, AUROC, TAR@FAR, and few-shot accuracy.

2.8 Strongest split for Q1

The strongest publishable protocol is a **protocol suite**, not one split:

1. In-distribution 25-person regular closed-set ID.
2. Cross-direction ID.
3. Cross-bag ID.
4. Combined cross-direction + cross-bag ID.
5. Few-shot enrollment / verification.
6. Cross-setup R → M and M → R on the 5 overlapping people.
7. Ablation and statistical tests across all protocols.

A paper that succeeds on this suite will be much stronger than a paper reporting only random-split accuracy.

PART 3 — Publishable improvement scope

3.1 Signal preprocessing improvements

Your baseline should include packet filtering, corrupted-packet removal, timestamp alignment, packet-rate normalization, and missing-packet handling. This matters because CSI has receiver gain and phase errors; recent preprocessing work explicitly models gain and synchronization-induced phase errors and argues that raw CSI can significantly hinder sensing tasks.

Q1 strength: medium by itself, high if linked to robustness under bag/direction/setup shifts.

Experiments: raw vs denoised vs calibrated; amplitude-only vs phase-only vs fused; all tested under hard splits.

3.2 CSI amplitude/phase cleaning

Amplitude is usually more stable; phase can carry spatial information but is sensitive to synchronization errors. Phase-processing work notes that CSI phase is underused because of synchronization errors and proposes sanitization plus smoothing/rebuilding pipelines.

Recommended: amplitude baseline first, then sanitized phase, then amplitude+phase fusion.

Risk: bad phase sanitization can reduce accuracy. Do not claim phase is always better; prove it.

3.3 Feature representation improvements

Use multiple representations:

- raw CSI amplitude tensor,
- PCA-reduced amplitude,
- sanitized phase,
- amplitude+phase fusion,
- Doppler/time-frequency spectrogram,
- CSI “image” for 2D CNN/ViT,
- tokenized CSI sequence for transformers,
- statistical vector for ML baselines.

Q1 strength: high if you show which representation generalizes best, not just which is best on random split.

3.4 Model architecture improvements

A publishable architecture should not be just “Transformer on CSI.” Better: **multi-branch robust representation learning**:

- temporal branch: TCN/BiGRU/Transformer,
- spectral/subcarrier branch: CNN or attention over subcarriers,
- condition-adversarial branch: remove direction/bag/setup nuisance,
- identity head: ArcFace/CosFace classification,
- auxiliary heads: direction and bag prediction,
- domain head: setup confusion for regular/modified alignment.

Q1 strength: high if evaluated against CNN, CNN-LSTM, BiGRU, TCN, Transformer, Wi-Gait-style residual Transformer, and metric-learning baselines.

3.5 Robustness and generalization improvements

Your dataset is ideal for robustness:

- train normal → test bag,
- train one direction → test opposite,

- train regular → test modified,
- train with/without augmentation,
- report degradation curves.

Q1 strength: high, because reviewers care more about robustness than one high random-split accuracy.

3.6 Cross-domain/cross-setup adaptation

Use the 5 overlapping subjects for:

- CORAL,
- MMD,
- DANN,
- domain-specific batch normalization,
- cross-attention alignment,
- test-time entropy minimization,
- few-shot target adaptation.

Q1 strength: medium with 5 subjects; high only if framed as a secondary validation or if you collect more modified-router participants.

3.7 Few-shot and low-data learning

Because every person has 100 samples, you can simulate 1-shot, 3-shot, 5-shot, 10-shot, 20-shot enrollment.

Use:

- prototypical networks,
- Siamese/triplet networks,
- supervised contrastive learning,
- ArcFace/CosFace,
- masked CSI pretraining + few-shot fine-tuning.

Q1 strength: high, because CAUTION and AutoFi show that low-label and few-shot WiFi sensing are active research directions.

3.8 Explainability

Use:

- subcarrier importance,
- time-region saliency,
- attention maps,
- UMAP/t-SNE embeddings by person/direction/bag/setup,
- confusion matrix ordered by physical similarity if metadata exists,
- ablation of antenna streams/subcarriers.

Q1 strength: medium to high if used to support physical interpretation, not just decorative figures.

3.9 Real-world deployment

Report:

- model parameters,
- FLOPs,
- inference latency,
- memory,
- window length,
- number of packets required,

- effect of reducing subcarriers,
- edge feasibility.

Q1 strength: medium by itself, high if targeting IEEE Internet of Things Journal, IEEE Sensors Journal, or Computer Networks.

3.10 Statistical reliability

Mandatory for Q1:

- 5 or 10 random seeds,
- mean ± standard deviation,
- 95% confidence intervals,
- paired Wilcoxon signed-rank test or paired t-test,
- McNemar test for paired classification comparisons,
- effect size,
- per-person results.

PART 4 — Models to train

4.1 Traditional ML baselines

Models	Input	Preprocessing	Loss/training	Metrics	Level
SVM, RF, XGBoost/LightGBM/CatBoost, KNN, Logistic Regression, LDA/QDA, Naive Bayes	Statistical features, PCA features, Doppler features	denoise, normalize, PCA/statistics	cross-entropy or tree objective	accuracy, macro-F1, balanced accuracy	mandatory baseline
HMM	sequence of gait-cycle features	segmentation/gait-cycle detection	likelihood maximization	sequence accuracy, EER	optional baseline

Use features such as mean/std/skew/kurtosis per subcarrier, Doppler-band energy, PCA component statistics, STFT energy, entropy, autocorrelation, and gait-cycle timing.

These models are not Q1 contributions, but reviewers expect them.

4.2 Classical deep learning

Model	Input	Strength	Weakness	Level
1D CNN	time × channel tensor	fast, strong baseline	limited long-range modeling	good baseline
2D CNN	time × subcarrier CSI image/spectrogram	strong spatial-temporal learning	sensitive to representation	good baseline
CNN-LSTM / CNN-GRU	CNN features + temporal sequence	strong for gait dynamics	slower, overfitting risk	strong baseline
BiLSTM / BiGRU	sequence	captures bidirectional patterns	not deployment-friendly if long windows	good
TCN	sequence	efficient long receptive field	needs tuning	very good
Autoencoder	reconstruction/pretraining	useful denoising	not enough alone	support method
Siamese/triplet network	paired/triplet samples	few-shot/verification	sampling sensitive	Q1-capable if well done

Use cross-entropy for closed-set ID; binary cross-entropy for verification; triplet/supervised contrastive loss for metric learning; reconstruction loss for autoencoders.

4.3 Attention and Transformer models

Model	Input	Use case	Q1 value
Vanilla Transformer Encoder	tokenized time windows	long-range temporal relations	good, but not novel alone
Temporal Transformer	packets as tokens	gait dynamics	strong baseline
Conformer	local CNN + global attention	CSI has local and long-range structure	strong candidate
ViT	CSI spectrogram/image patches	image-like representation	good if compared fairly
Swin Transformer	local shifted patches	large CSI images	risky with small data
Cross-attention domain model	regular and modified features	setup adaptation	Q1-level if done carefully
Informer/Autoformer/FEDformer	long time series	only useful for long windows	probably overkill

A strong model for your case:

CSI-RobustFormer

- branch 1: temporal convolution,
- branch 2: subcarrier attention,
- branch 3: masked CSI reconstruction pretraining,
- identity head with ArcFace,
- auxiliary direction/bag heads,
- domain-adversarial setup head for regular/modified.

4.4 Self-supervised and contrastive learning

Method	Input	Why useful	Level
SimCLR-style CSI contrastive	augmented windows	identity-invariant representation	Q1-capable
MoCo	memory queue embeddings	stable with small batch	Q1-capable
BYOL / Barlow Twins / VICReg	two augmented views	avoids negative sampling issues	strong
TS2Vec	time-series representation	good for window-level CSI	strong
CPC	predict future CSI features	gait temporal structure	strong
Masked CSI modeling	mask packets/subcarriers	transformer pretraining	very strong
Denoising autoencoder	noisy → clean CSI	preprocessing/model bridge	support

Good augmentations:

- time crop,
- packet dropout,
- subcarrier dropout,
- Gaussian jitter,
- amplitude scaling,
- time warping,
- frequency masking,
- mild phase perturbation after sanitization,
- direction-aware reversal only for training augmentation.

Do not use augmentations that destroy identity-related gait dynamics.

4.5 Domain adaptation/generalization

Method	Use	Target protocol
CORAL	align covariance	regular → modified
MMD	distribution alignment	regular → modified
DANN	adversarial domain-invariant features	setup/domain adaptation
Domain-specific batch norm	preserve setup-specific statistics	regular and modified
Test-time adaptation	adapt to modified without labels	realistic deployment
Meta-learning	simulate setup/condition shifts	few-shot setup adaptation
Cross-attention alignment	align class-level regular/modified embeddings	strongest novelty candidate

Because modified setup has only 5 people, use this as a **secondary external-domain test**, not as your entire paper.

4.6 Few-shot and metric learning

Use:

- Prototypical Networks,
- Matching Networks,
- Relation Networks,
- Siamese network,
- triplet network,
- supervised contrastive learning,
- ArcFace/CosFace/SphereFace.

Best combination:

Self-supervised pretraining → ArcFace supervised fine-tuning → prototypical few-shot evaluation.

Report 1-shot, 3-shot, 5-shot, 10-shot, and 20-shot results per person.

4.7 LLM-inspired approaches

A pure LLM is **not scientifically justified** as the core model for CSI gait recognition. CSI is numeric RF time-series data, not language. Using an LLM just because it is trendy will look weak.

LLMs can help with:

- experiment planning,
- automated result summarization,
- code generation,
- error-log interpretation,
- converting metadata into natural-language documentation,
- generating manuscript drafts.

Better scientific alternatives are:

- transformer encoders,
- masked CSI modeling,
- time-series foundation-style pretraining,
- contrastive representation learning,
- self-supervised domain adaptation.

PART 5 — Dataset preparation pipeline

5.1 Folder organization

```
csi_gait_project/
  data/
    raw/
   regular/
    P001/
     csi/
     csv/
    P002/
    ...
   modified/
    P001/
     csi/
     csv/
    ...
    processed/
   amplitude/
   phase_sanitized/
   amp_phase_fusion/
   spectrogram/
   pca_features/
   statistical_features/
    metadata/
   metadata_master.csv
   participant_info_anonymized.csv
   hardware_setup.yaml
   collection_protocol.md
   file_checksums.csv
  splits/
   random_split_seed42.csv
   cross_direction_L2R_to_R2L.csv
   cross_direction_R2L_to_L2R.csv
   cross_bag_normal_to_bag.csv
   cross_setup_regular_to_modified.csv
   fewshot_protocols/
    scripts/
   01_build_metadata.py
   02_validate_files.py
   03_extract_csi.py
   04_preprocess_csi.py
   05_make_splits.py
   06_train_baselines.py
   07_train_deep.py
   08_eval_protocols.py
    experiments/
   configs/
   logs/
   checkpoints/
   tensorboard/
   wandb/
    results/
   tables/
   figures/
   confusion_matrices/
```

```
  embeddings/
  statistical_tests/
    manuscript/
  paper1/
  paper2/
```

5.2 Metadata table

Minimum metadata_master.csv:

```
participant_id
sample_id
setup_type              # regular / modified
direction               # left_to_right / right_to_left
condition               # normal / with_bag
csi_file_path
csv_file_path
collection_date
session_id
router_config
antenna_config
antenna_positions
receiver_config
ap_receiver_distance_m
walking_path_length_m
walking_path_coordinates
sampling_rate_hz
num_packets
num_subcarriers
num_tx
num_rx
band
channel
bandwidth_mhz
mcs
nss
packet_loss_rate
duration_s
environment_id
room_length_m
room_width_m
line_of_sight
bag_weight_kg
notes
checksum
```

5.3 CSI preprocessing pipeline

Recommended pipeline:

1. Load PicoScenes CSI.
2. Validate packet timestamps.
3. Remove corrupted packets.
4. Remove abnormal RSSI/CSI amplitude outliers.
5. Align or resample packet sequence.
6. Extract complex CSI.
7. Extract amplitude and phase.

8. Apply amplitude denoising: Hampel filter, median filter, low-pass filter.
9. Apply phase unwrapping and sanitization.
10. Remove static component if needed.
11. PCA across subcarriers/antennas.
12. Generate STFT/spectrogram/Doppler map.
13. Normalize using train-set statistics only.
14. Segment into fixed windows.
15. Save processed tensors and metadata.
16. Assign labels.
17. Apply augmentation only inside training pipeline.

5.4 Model-ready formats

Format	Shape idea	Suitable models
Raw complex CSI	packets × subcarriers × antennas × 2	advanced fusion models
Amplitude tensor	packets × subcarriers × links	1D CNN, TCN, Transformer
Phase tensor	packets × subcarriers × links	phase-specific models
Amplitude+phase	packets × subcarriers × links × 2	CNN/Transformer fusion
Spectrogram	frequency × time × channels	2D CNN, ViT, Swin
PCA features	packets × components	ML, LSTM, TCN
Statistical vector	feature vector	SVM/RF/XGBoost
Token sequence	tokens × embedding	Transformer/Conformer

PART 6 — Q1 experimental design

6.1 Core experiments

Experiment	Data	Train	Test	Purpose
E1 random closed-set	25 regular	stratified random	stratified random	easy baseline
E2 grouped sample split	25 regular	early/middle samples	held-out samples	leakage-reduced baseline
E3 cross-direction	25 regular	L2R	R2L	direction robustness
E4 reverse cross-direction	25 regular	R2L	L2R	symmetry check
E5 cross-bag	25 regular	normal	bag	carrying robustness
E6 combined hard	25 regular	normal L2R	bag R2L	hardest condition shift
E7 bag classification	25 regular	mixed persons	normal vs bag	auxiliary task
E8 direction classification	25 regular	mixed persons	L2R vs R2L	auxiliary task
E9 multitask	25 regular	ID + direction + bag	all test protocols	representation learning
E10 few-shot ID	25 regular	k samples/person	remaining	enrollment efficiency
E11 verification	25 regular	enrollment/query pairs	genuine/impostor	biometric metric
E12 R → M	5 overlap	regular	modified	setup shift
E13 M → R	5 overlap	modified	regular	reverse shift
E14 DA R → M	5 overlap	labeled R + unlabeled M	M	adaptation
E15 ablation	all	controlled	controlled	contribution proof

6.2 Mandatory ablations

- amplitude only vs phase only vs amplitude+phase,
- raw vs filtered vs PCA vs spectrogram,
- no augmentation vs direction-aware augmentation,
- CNN vs CNN-GRU vs TCN vs Transformer vs proposed model,
- cross-entropy vs ArcFace vs supervised contrastive,
- single-task vs multi-task,
- no domain adaptation vs CORAL/MMD/DANN/cross-attention,
- regular-only vs modified-only vs combined,
- different numbers of training samples per person.

6.3 Statistical validation

For every major table:

```
mean ± std over 5 or 10 seeds
95% confidence interval
paired statistical test against strongest baseline
per-person accuracy/F1
macro-F1 and balanced accuracy
confusion matrix
EER/AUROC for verification
```

6.4 Complexity analysis

Report:

- parameters,
- FLOPs,
- inference time per sample/window,

- memory,
- preprocessing time,
- minimum packets/window needed,
- performance vs window length,
- performance vs number of subcarriers.

6.5 Explainability

Best explainability figures:

- time saliency over walking sequence,
- subcarrier importance,
- antenna-link importance,
- attention heatmap,
- UMAP colored by person/direction/bag/setup,
- before/after domain adaptation embedding plot,
- failure-case visualization for confused identities.

PART 7 — Serious Q1 paper ideas

Idea 1 — Robust CSI gait recognition under direction and carrying-condition shifts

Tentative title:
"Direction- and Carrying-Condition-Robust WiFi CSI Gait Recognition via Multi-Task Contrastive Representation Learning"

Problem: Person recognition degrades when walking direction or carrying condition changes.

Novelty: Jointly learns identity, direction, and carrying-condition representations while using contrastive/ArcFace losses to preserve identity and reduce nuisance sensitivity.

Dataset: 25-person regular-router dataset.

Architecture: CSI encoder with CNN/TCN/Transformer backbone; identity head; direction head; bag head; supervised contrastive or ArcFace loss.

Baselines: SVM, RF, XGBoost, CNN, CNN-LSTM, BiGRU, TCN, Transformer, Wi-Gait-style residual Transformer, Siamese/triplet.

Experiments: random, cross-direction, cross-bag, combined hard split, ablations, explainability.

Target journals: IEEE Internet of Things Journal, IEEE Sensors Journal, IEEE Transactions on Instrumentation and Measurement, Computer Networks, Expert Systems with Applications.

Risk: If only random accuracy is high, paper is weak. The hard-split results must be convincing.

Idea 2 — Self-supervised masked CSI modeling for few-shot gait/person recognition

Tentative title:
"Masked CSI Modeling for Few-Shot WiFi Gait-Based Person Recognition"

Problem: WiFi gait models usually need many labeled samples per person.

Novelty: Pretrain a transformer/Conformer by reconstructing masked packets/subcarriers, then fine-tune with few labels.

Dataset: All regular samples; modified setup as external-domain stress test.

Architecture: Masked CSI transformer + ArcFace/prototypical classifier.

Baselines: supervised CNN, CNN-GRU, TCN, vanilla Transformer, SimCLR, TS2Vec, CAUTION-style few-shot profile.

Experiments: 1/3/5/10/20-shot, cross-direction, cross-bag, R → M.

Target journals: IEEE Internet of Things Journal, IEEE TCCN, Information Fusion, Expert Systems with Applications.

Risk: Needs careful augmentation/pretraining design. If performance gain is small, novelty weakens.

Idea 3 — Antenna-spread router configuration and cross-setup adaptation

Tentative title:
"From Compact to Antenna-Spread WiFi CSI Gait Sensing: A Cross-Setup Domain Adaptation Study"

Problem: Router antenna placement changes CSI distributions and model generalization.

Novelty: Uses your physical antenna-spread router configuration as an explicit domain shift and evaluates adaptation.

Dataset: 5 overlapping participants, regular and modified; optionally all 25 regular for source pretraining.

Architecture: shared encoder + domain-adversarial head + class-level MMD/CORAL/cross-attention alignment.

Baselines: no adaptation, fine-tuning, CORAL, MMD, DANN, domain-specific BN.

Experiments: R → M, M → R, few-shot target adaptation, embedding alignment, per-subject comparison.

Target journals: Computer Networks, Ad Hoc Networks, IEEE Sensors Journal, IEEE Internet of Things Journal.

Risk: 5 people is not enough for a standalone strong Q1 claim. Treat this as a second contribution or collect more modified-router participants.

Idea 4 — Lightweight deployable CSI gait recognition

Tentative title:
"EdgeCSI-Gait: Lightweight WiFi CSI Person Recognition with Robustness-Aware Compression"

Problem: Deep CSI models are often too heavy for practical deployment.

Novelty: TCN/mobile CNN model with pruning/quantization and robustness-preserving training.

Dataset: 25 regular + 5 modified external test.

Architecture: depthwise separable CNN + TCN + ArcFace; optional quantization-aware training.

Baselines: full CNN, LSTM, Transformer, TCN.

Experiments: accuracy vs latency vs FLOPs; cross-bag/direction; model compression.

Target journals: IEEE Sensors Journal, IEEE Internet of Things Journal, Computer Networks.

Risk: Must not become a simple compression paper. Tie deployment to hard robustness protocols.

Idea 5 — Metric-learning CSI biometric verification

Tentative title:
"Open-Set WiFi CSI Gait Biometrics with Metric Learning and Carrying-Condition Stress Testing"

Problem: Closed-set accuracy is unrealistic; real systems need enrollment and impostor rejection.

Novelty: Converts your dataset into biometric verification/open-set protocols with EER, FAR/FRR, and few-shot enrollment.

Dataset: 25 regular; 5 modified as external domain.

Architecture: Siamese/triplet/prototypical/ArcFace encoder.

Baselines: softmax classifier, SVM, CAUTION-style profile threshold, triplet CNN.

Experiments: known/unknown identities, 1–20 shot enrollment, normal → bag, direction shift, R → M.

Target journals: IEEE Transactions on Information Forensics and Security, IEEE Internet of Things Journal, Pattern Recognition, Information Fusion.

Risk: 25 subjects may be marginal for TIFS/Pattern Recognition unless experiments are very rigorous.

Idea 6 — Multi-view CSI representation learning

Tentative title:
"Multi-View Amplitude–Phase–Doppler Fusion for Robust WiFi CSI Gait Recognition"

Problem: Existing models often use only amplitude or only local time-frequency features.

Novelty: Fuse amplitude, sanitized phase, and Doppler/spectrogram views with view-consistency contrastive learning.

Dataset: 25 regular; all protocols.

Architecture: three encoders + cross-view attention + contrastive alignment.

Baselines: amplitude-only, phase-only, spectrogram-only, early fusion, late fusion.

Risk: Must prove phase cleaning is valid and not noisy.

Idea 7 — Physical interpretation of antenna-spread CSI gait sensing

Tentative title:
"Does Spatially Spreading Router Antennas Improve CSI-Based Gait Discriminability?"

Problem: Low-cost router antenna placement is rarely analyzed as a sensing-design variable.

Novelty: Physical analysis of discriminability, SNR, subcarrier saliency, embedding separation, and cross-setup shift.

Dataset: 5 overlap only.

Risk: Not enough for standalone Q1 unless expanded. Better as a section in Paper 2.

PART 8 — Best two-paper strategy

Paper 1 — safest Q1 paper

Central idea:
Robust WiFi CSI person recognition under direction and carrying-condition shifts.

Use dataset:
Mainly 25-person regular-router dataset.

Models:
Traditional ML, CNN, CNN-GRU, TCN, Transformer, proposed multi-task contrastive/ArcFace CSI encoder.

Novelty:
Not "new model only." The novelty should be:

1. leakage-safe CSI gait evaluation,
2. hard cross-direction and cross-bag protocols,
3. multi-task/metric representation that preserves identity while handling nuisance factors,
4. explainable subcarrier/time-region analysis.

Minimum experiments:
E1–E8, E10–E11, ablations, statistical tests, explainability.

Suitable journals:
IEEE Internet of Things Journal, IEEE Sensors Journal, IEEE Transactions on Instrumentation and Measurement, Computer Networks, Expert Systems with Applications.

Acceptance chance:
Highest among your options, provided results are honest and hard-split performance is strong.

Paper 2 — stronger novelty but higher risk

Central idea:
Cross-setup/domain-adaptive CSI gait recognition between compact-router and antenna-spread-router configurations.

Use dataset:
5 overlapping subjects in regular and modified setups; use all 25 regular subjects for representation pretraining but evaluate cross-setup only on the 5 overlaps.

Models:
CORAL, MMD, DANN, domain-specific BN, test-time adaptation, cross-attention alignment, masked CSI pretraining.

Novelty:
Low-cost physical antenna-spread configuration as a sensing-domain intervention + paired cross-setup adaptation.

Minimum experiments:
R → M, M → R, no adaptation vs adaptation, few-shot target adaptation, embedding alignment, setup-specific saliency, antenna-link/subcarrier importance, statistical testing.

Suitable journals:
Computer Networks, Ad Hoc Networks, IEEE Sensors Journal, IEEE Internet of Things Journal. For IEEE TIFS or Pattern Recognition, you likely need more modified-router subjects.

Acceptance chance:
Moderate if framed carefully; weak if claimed as a general antenna-design breakthrough.

Which paper first?

Write **Paper 1 first**. It uses your strongest portion: 25 subjects, 2,500 regular samples, direction labels, bag labels.

Paper 2 requires more careful framing and is more vulnerable because of the 5-person modified subset. If possible, collect modified-router data for at least 10–15 more of the existing 25 participants **before submitting Paper 2.**

PART 9 — Step-by-step roadmap

Stage	Goal	Exact tasks	Tools	Output	Common mistakes	QC checklist
1. Dataset audit	Verify integrity	count files, match CSI/CSV pairs, detect missing/corrupt samples	`Python, pandas, pathlib`	audit report	training before audit	3,000 expected sample pairs if complete
2. Metadata	Build master table	parse IDs, setup, sample index, labels	`pandas`	metadata_master.csv	manual labels	all labels deterministic
3. CSI loading	Convert raw CSI	load PicoScenes files, export tensors	PicoScenes tools, CSIKit/csiread if compatible	raw tensor cache	assuming fixed dimensions	log packets/subcarriers per file
4. Preprocessing	Clean CSI	filter, outlier removal, phase sanitization, normalization	`NumPy, SciPy, PyWavelets`	processed tensors	fitting scaler on all data	train-only normalization
5. Label generation	Build labels	person, direction, bag, setup	`pandas`	label columns	off-by-one index	verify odd/even and 1–80/81–100
6. EDA	Understand signals	duration, packet count, PCA, spectrogram examples	matplotlib, UMAP	EDA report	only showing pretty plots	include failures/outliers
7. ML baselines	Establish floor	SVM/RF/XGB/KNN	scikit-learn, XGBoost	baseline table	using random split only	all hard protocols
8. DL baselines	Establish fair comparisons	CNN, CNN-GRU, TCN, Transformer	PyTorch	baseline DL results	overfitting small data	same splits/seeds
9. Advanced models	Build contribution	multi-task, contrastive, masked CSI, DA	PyTorch Lightning	proposed results	too many models with no story	one main model, many baselines
10. Ablations	Prove components	remove heads/losses/views	scripts	ablation table	ablation only random split	ablate under hard split
11. Statistics	Validate claims	CI, tests, per-person stats	scipy, statsmodels	statistical report	single-run claims	5–10 seeds
12. Explainability	Interpret model	saliency, attention, subcarrier importance	Captum, UMAP	figures	decorative heatmaps	connect to physical setup
13. Manuscript	Write paper	intro, related work, methods, results	LaTeX/Overleaf	draft	overstated claims	reviewer checklist
14. Journal selection	Match scope	choose journal based on contribution	journal finder, Scopus/SJR check	target list	chasing highest journal only	scope-fit first
15. Submission	Final package	code, splits, metadata, supplementary	GitHub/Zenodo if possible	final submission	missing reproducibility	seeds, configs, splits released

PART 10 — Manuscript guidance

10.1 Strong title options

For Paper 1:

1. **Direction- and Carrying-Condition-Robust WiFi CSI Gait Recognition via Multi-Task Contrastive Learning**
2. **Robust Device-Free Person Identification from WiFi CSI Under Walking Direction and Bag-Carrying Variations**
3. **Leakage-Safe Evaluation and Robust Representation Learning for WiFi CSI Gait Biometrics**

For Paper 2:

1. **Cross-Setup WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Configurations**
2. **Domain-Adaptive WiFi CSI Gait Recognition with Antenna-Spread Spatial Sensing**
3. **A Paired Study of Router Antenna Configuration Shift in WiFi CSI-Based Person Recognition**

10.2 Abstract structure

Use this structure:

1. Problem: WiFi CSI gait recognition suffers under direction, carrying-condition, and setup shifts.
2. Gap: most works report high accuracy under easier splits and limited covariate stress testing.
3. Dataset: 25 participants, 100 samples/person, direction labels, bag labels, 5 paired regular/modified subjects.
4. Method: robust CSI preprocessing + multi-task/contrastive/domain-adaptive model.
5. Protocols: random, cross-direction, cross-bag, cross-setup, few-shot/open-set.
6. Results: report actual numbers only after experiments.
7. Impact: robust, reproducible, device-free CSI gait recognition.

10.3 Best figures

Mandatory figures:

1. hardware setup diagram,
2. regular router setup geometry,
3. modified antenna-spread setup geometry,
4. dataset hierarchy and label rules,
5. CSI preprocessing pipeline,
6. model architecture,
7. experimental protocol diagram,
8. representative CSI amplitude/spectrogram for direction and bag,
9. confusion matrix,
10. UMAP/t-SNE embedding before and after proposed method,
11. saliency/subcarrier importance,
12. accuracy degradation across protocols,
13. cross-setup domain adaptation embedding.

10.4 Best tables

Mandatory tables:

1. comparison with existing CSI gait/person-ID works,
2. dataset statistics,
3. metadata summary,
4. split protocols,

5. baseline results,
6. proposed model vs baselines,
7. cross-direction/cross-bag results,
8. cross-setup results,
9. ablation study,
10. complexity analysis,
11. statistical significance.

10.5 How to present the modified-router setup

Do not say:

> "We created a new 4 m MIMO antenna array."

Say:

> "We physically separated two 5 GHz router antennas by approximately 4 m to create an antenna-spread router configuration that changes the spatial illumination and multipath sensitivity of the walking region."

Also state:

> "The configuration is not claimed to be a calibrated antenna array; it is treated as a practical router-configuration domain shift."

That wording is much safer.

PART 11 — Reviewer-style criticism

11.1 Weaknesses that may cause rejection

1. **Only one environment**: weak real-world generalization.
2. **Only 5 modified-router subjects**: too small for standalone antenna-spread claims.
3. **No session metadata**: reviewers may suspect temporal leakage.
4. **Random split only**: likely rejection for serious Q1.
5. **No strong baselines**: Transformer-only papers look incremental.
6. **No statistical testing**: single-run accuracy is not enough.
7. **No public splits/code**: reproducibility criticism.
8. **No biometric verification metrics**: person recognition should not rely only on closed-set accuracy.
9. **Overclaiming privacy/security**: WiFi CSI can still leak private attributes; avoid absolute privacy claims.
10. **No environmental variation**: cannot claim deployment-ready generalization.

11.2 Claims to avoid

Avoid:

- “universal gait biometric system,”
- “real-world robust” unless tested in multiple environments,
- “privacy-preserving” without nuance,
- “antenna-spread router solves CSI gait recognition,”
- “Q1-level dataset” as a claim,
- “LLM-based CSI recognition” unless scientifically justified,
- “phase improves performance” unless phase cleaning and ablations prove it.

11.3 Mandatory experiments

For Paper 1:

- random split,
- leakage-safe grouped split,
- cross-direction,
- cross-bag,
- combined cross-direction + cross-bag,
- few-shot,
- verification/open-set,
- amplitude/phase/spectrogram ablation,
- model ablation,
- statistical testing.

For Paper 2:

- regular → modified,
- modified → regular,
- unsupervised adaptation,
- few-shot target adaptation,
- no-adaptation baseline,
- embedding/domain visualization,
- per-subject results.

11.4 Are 25 participants enough?

Yes, **25 participants can be enough** for a serious Q1 paper if the contribution is robustness, representation learning, leakage-safe evaluation, and careful modeling.

No, 25 participants are **not enough** to claim a deployable general biometric system across populations/environments.

11.5 Is the 5-person modified subset enough?

Not for a standalone high-impact Q1 claim.

It is enough for:

- pilot cross-setup validation,
- paired domain-shift analysis,
- secondary experiment,
- proof-of-concept for antenna-spread router configuration.

For a strong standalone Paper 2, collect modified-router samples for at least **15–20 participants**, ideally all 25.

11.6 What makes the paper look incremental?

- CNN/LSTM/Transformer comparison only.
- Random split accuracy only.
- No cross-condition testing.
- No domain adaptation despite having modified setup.
- No leakage discussion.
- No metadata.
- No statistical tests.
- No release of split files.

11.7 What makes it look serious?

- Hard protocols.
- Honest performance drops.
- Strong baselines.
- Few-shot/open-set biometric evaluation.
- Domain-shift analysis.
- Statistical testing.
- Explainability tied to physical CSI behavior.
- Clear metadata and reproducible splits.

PART 12 — Final deliverables

12.1 Ranked research ideas

1. **Direction- and bag-robust CSI gait recognition with multi-task contrastive learning** — safest and strongest.
2. **Masked/self-supervised CSI pretraining for few-shot person recognition** — very publishable if done well.
3. **Open-set/verification CSI gait biometrics with metric learning** — strong biometric framing.
4. **Cross-setup regular ⇄ modified antenna-spread domain adaptation** — novel but limited by 5 subjects.
5. **Lightweight deployable CSI gait model** — good secondary angle.
6. **Multi-view amplitude–phase–Doppler fusion** — good if phase quality is strong.
7. **Antenna-spread physical analysis alone** — interesting, but needs more modified data.

12.2 Ranked models to train first

1. SVM/RF/XGBoost on statistical + PCA features.
2. 1D CNN amplitude baseline.
3. CNN-GRU or CNN-BiGRU.
4. TCN.
5. Transformer encoder.
6. ArcFace/CosFace identity model.
7. Supervised contrastive model.
8. Multi-task CSI encoder: ID + direction + bag.
9. Masked CSI pretraining model.
10. Domain adaptation model for regular ⇄ modified.

12.3 Full experiment matrix

Phase	Experiment	Models
Baseline	random 25-person ID	ML, CNN, GRU, TCN
Baseline	grouped 25-person ID	ML, CNN, GRU, TCN
Robustness	L2R → R2L	CNN, TCN, Transformer, proposed
Robustness	R2L → L2R	CNN, TCN, Transformer, proposed
Robustness	normal → bag	CNN, TCN, Transformer, proposed
Robustness	normal L2R → bag R2L	proposed + strongest baselines
Auxiliary	direction classification	CNN/TCN
Auxiliary	bag classification	CNN/TCN
Multi-task	ID + direction + bag	proposed
Few-shot	1/3/5/10/20-shot	ProtoNet, ArcFace, contrastive
Verification	genuine/impostor	Siamese, triplet, ArcFace
Cross-setup	regular → modified	no adaptation, CORAL, MMD, DANN
Cross-setup	modified → regular	same
Ablation	views/losses/heads	proposed variants
Deployment	latency/FLOPs/window length	best compact models

12.4 Recommended timeline

Month	Work
1	dataset audit, metadata, preprocessing, split files
2	EDA, ML baselines, CNN/GRU/TCN baselines
3	Transformer, ArcFace, contrastive learning
4	multi-task model, cross-direction/cross-bag experiments
5	few-shot/open-set experiments, ablations
6	statistical tests, explainability, Paper 1 writing
7	Paper 1 submission; begin cross-setup models
8	R → M/M → R adaptation, target few-shot experiments
9	Paper 2 draft; collect more modified data if possible
10	Paper 2 submission

12.5 Publication strategy

Submit Paper 1 first to IEEE Internet of Things Journal, IEEE Sensors Journal, IEEE Transactions on Instrumentation and Measurement, Computer Networks, or Expert Systems with Applications, depending on final novelty and results.

Submit Paper 2 second only after you either:

1. collect more modified-router participants, or
2. frame it as a cross-setup adaptation study with the 5-person subset clearly labeled as a paired pilot domain.

12.6 Q1 readiness checklist

Before submission, you should have:

- complete metadata table,
- file integrity report,
- reproducible split files,
- leakage-safe protocol description,
- at least 6 strong baselines,
- cross-direction results,
- cross-bag results,
- few-shot or verification results,
- statistical tests,
- ablation study,
- complexity analysis,
- explainability figures,
- honest limitations section,
- code/config/split release plan,
- no overclaiming about real-world deployment.

Final best recommended plan

Your best route is:

First: build a robust 25-person regular-router paper around **direction and bag-condition robustness** using multi-task contrastive/ArcFace representation learning. This is the safest Q1 paper.

Second: use the modified-router subset for a **cross-setup/domain-adaptation paper**, but either collect more modified-router data or frame it conservatively as a paired cross-configuration study.

Do not start with a data journal. Do not start with the 5-person modified setup alone. Do not rely on random-split accuracy.

Start with metadata, leakage-safe splits, amplitude baselines, and hard protocols. Then move to the proposed model. The paper will be taken seriously if it shows not only high accuracy, but also where CSI gait recognition fails and how your method reduces those failures.